

Causally interpretable meta-analysis combining aggregate and individual participant data

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Abstract

Recent work in causally interpretable meta-analysis (CIMA) has bridged the gap between traditional meta-analysis and causal inference. Whereas traditional meta-analysis results generally do not apply to any well-defined population, CIMA approaches specify a target population to which meta-analytic treatment effect estimates are transported. Although theoretically attractive, these approaches currently have some practical limitations. Most assume that all studies in the meta-analysis have individual participant data (IPD), which is rare in practice because most trials share only aggregate data. We propose a method to perform CIMA using a combination of aggregate data and IPD. This method borrows information from studies with IPD to augment the aggregate data and create aggregate-matched synthetic IPD (AMSIPD), which can be used readily in the existing CIMA framework. By allowing use of both aggregate data and IPD, the method opens CIMA to more applications and can avoid biases arising from using only studies with IPD. We present a case study and simulations showing the AMSIPD approach is promising and merits further investigation as an advancement of CIMA.

Key words: meta-analysis; causal inference; weighting; IPD; transportability.

Introduction

Randomized controlled trials (RCTs) are considered the highest evidence level,¹ especially when assessing a treatment's causal effect, because randomization ensures high internal validity. The estimated causal treatment effect is presumed unbiased for the population sampled in the study, because potential confounders have the same distribution, on average, in each of the study's treatment groups.² Not surprisingly, systematic reviews and meta-analyses combining results of multiple RCTs are considered the highest-quality body of evidence.³ If an RCT provides internally valid evidence, a collection of RCTs is assumed to be better, providing a larger sample size and diverse study populations.

Although RCTs and related meta-analyses have high internal validity, their external validity is often questioned. A trial's eligibility criteria and convenience sampling create study samples that may not closely resemble a population of interest. Practitioners are often interested in a treatment's effect in a population different than the study populations of extant RCTs or the union of study populations in a meta-analysis. Transporting or generalizing results to a specific target population is, therefore, of great interest.

Significant effort has been made to address these external validity issues. The vast generalizability and transportability literature focuses on using results from RCTs to estimate a treatment effect in a target population, from which a sample is available. These methods generally assume individual participant data (IPD) are available for the trial and the target population sample. They also assume that differences between the populations modify the treatment effect and that adjusting for effect modifiers gives

a more accurate estimate of the treatment effect in the target population. For example, Shao et al.³ found a different effect of dapagliflozin on heart failure hospitalizations in patients with diabetes depending on whether they had atherosclerotic cardiovascular disease. The goal of “transporting” an estimated treatment effect is to account for such differences between the trial and target populations rather than assuming the populations are similar enough that the differences can be ignored.

Although less transportability research has been done for meta-analyses, it is a field with recent and growing activity. For this article, we relied particularly on the work of Dahabreh et al.,^{4,5} which extends to meta-analysis concepts from RCT transportability and generalizability. This extension may seem obvious but is a paradigm shift in meta-analysis methods and interpretation. Most traditional meta-analysis methods estimate treatment effects using a weighted average of study-specific treatment effect estimates, often weighting by the precision of those estimates. Weighting by precision gives the minimum-variance estimate within a certain class by weighting trials with larger sample sizes more than smaller trials, but trial sample sizes are determined by funding and other factors unrelated to the trial populations' similarity to a target population. When transporting estimated treatment effects to a specific target population, weighting by precision is not only unfounded but potentially harmful. Thus, causal methods like those of Dahabreh et al.^{4,5} are an important development in meta-analysis. Whereas “traditional meta-analysis estimators estimate the effect in a population that is at best nebulous and even potentially unknown,”⁶ the new causally interpretable meta-analysis (CIMA) estimates a treatment effect in a specific target population.

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The approach of Dahabreh et al.^{4,5} follows the above description of transportability of RCTs. It assumes IPD are available for the meta-analysis studies and for a sample from the target population. Using an outcome model or inverse probability weighting, or a so-called doubly robust approach combining both, the meta-analysis studies are transported to the target population. The estimated average treatment effect (ATE) in the target population thus accounts for effect modification, assuming the relevant effect modifiers have been collected. Although CIMA is an attractive theoretical development, these methods have received little practical investigation.

Although IPD are increasingly available, few trials report IPD,⁷ and the current CIMA approach's IPD requirements are impractical. Commonly, investigators can access a combination of studies with IPD and aggregate data.

Excluding trials without IPD from a CIMA can produce bias if those trials differ systematically from IPD trials. For example, trials with smaller estimated treatment effects may be less likely to share IPD; omitting them will bias the causal treatment effect estimate.⁸ Such exclusion is also statistically inefficient, due to a smaller sample size.

Here, we propose a new method that allows the CIMA framework to be used for such a combination by borrowing information from the IPD studies to augment the aggregate data. Our method creates aggregate-matched synthetic IPD (AMSIPD) for individuals in aggregate-only studies, so that all studies have IPD. Aggregate matching ensures that the synthetic data reflect available aggregate information. Aggregate-matched synthetic IPD bears similarities to matching-adjusted indirect comparison,⁹ which, among other differences, does not focus on transportability.

Setup and assumptions

Assume each RCT has either IPD or aggregate data (eg, means, SDs, sample sizes for treatment groups), with at least 1 trial of each kind. Index the RCTs as $S = 1, 2, \dots, m$ from the set $\mathcal{S}$; IPD trials and aggregate-data trials belong to subsets $\mathcal{S}^I$ and $\mathcal{S}^{-I}$, respectively. The sample size in trial $S = s$ is n_s . Individual-level trial data include the trial label S , treatment A , covariates X , and outcome Y . We also have a random sample of IPD from the target population, labeled $S = 0$, with sample size n_0 , for which we require only the covariates X . Thus, for individual i , the observed data tuple is

$$O_i = \begin{cases} (X_i, S_i = 0), & \text{if } S_i = 0, \\ (X_i, A_i, Y_i, S_i), & \text{if } S_i \in \mathcal{S}^I, \\ S_i, (\bar{A}_S, \bar{X}_S^A, \bar{Y}_S^A) & \text{if } S_i \in \mathcal{S}^{-I}, \end{cases}$$

where over-bars indicate means but, in an abuse of notation, can include other summaries, such as SDs. For individuals in aggregate-only studies, we know their study S_i , the number of people in each treatment group in S_i , and the treatment-level covariate, and outcome means and SDs ($\bar{X}_S^A$ and $\bar{Y}_S^A$, where the superscript denotes a specific treatment arm). These assumptions are much more inclusive than requiring all studies to have IPD.

Because the method builds upon the CIMA framework of Dahabreh et al.,^{4,5} it requires the same assumptions, as follows. Suppose we have 2 treatments a and a' , with treatment assignment A . In the potential outcomes framework, Y^a denotes the potential (possibly counterfactual) outcome under treatment assignment $A \in \{a, a'\}$. We estimate the ATE in the target population, $E[Y^a - Y^{a'} | S = 0]$, using the identifiability conditions in

the following list,^{4,5} in which $f(\cdot)$ generically denotes appropriate densities.

1. *Consistency of potential outcomes.* When $A_i = a$, $Y_i = Y_i^a$ for all individuals in the target population or the populations forming $\mathcal{S}$. For example, there are no Hawthorne effects.¹⁰
2. *Conditional exchangeability over treatment assignment A .* $Y^a \perp\!\!\!\perp A \mid X = x, S = s$ for all trials $s \in \mathcal{S}$, each treatment $a \in \{a, a'\}$, and all x with $f(x, S = s) > 0$. The potential outcome Y^a is conceived as an innate characteristic existing before treatment assignment, so randomization implies this assumption.¹¹
3. *Positivity of treatment assignment probability in the meta-analysis studies.* For each treatment $a \in \{a, a'\}$, $\Pr[A = a | X = x, S = s] > 0$ for each study $s \in \mathcal{S}$ and all x with $f(x, S = s) > 0$. That is, all individuals in a study's covariate space could be assigned to either treatment. Randomization guarantees this.
4. *Conditional exchangeability in mean between study and target population.* For treatments a and a' , $E[Y^a - Y^{a'} | X = x, S = 0] = E[Y^a - Y^{a'} | X = x, S = s]$ for $s \in \mathcal{S}$ and all x with $f(x, S = 0) > 0$. This means trial participation does not affect the conditional ATE. This assumption is key to transporting effects from a study to the target population; Dahabreh et al.⁴ use a different form of this assumption, which affects the transportation.
5. *Positivity of trial participation probability.* $\Pr[S = s | X = x] > 0$ for $s \in \mathcal{S}$ and all x with $f(x, S = 0) > 0$. That is, each covariate pattern in the target population can occur in each meta-analysis study. This assumes the target population's covariate distribution overlaps that of the meta-analysis studies; our case study and many applications may violate this.

Under these assumptions, the studies can be pooled. In particular, assumption 4 implies that all between-study heterogeneity of treatment effects is captured by conditioning on covariates. Moreover, these assumptions imply there are no omitted between-study confounders and thus no omitted variable bias.

We further assume the target population sample and each meta-analysis study include the same covariates. This strong assumption has been addressed by Steingrimsdottir et al.¹² using assumptions and methods similar to those we use in the present article. We discuss this more later in this article.

To combine studies with aggregate-only and IPD data, we need further assumptions that we recognize are strong. Suppose individual i in study S has covariate vector X_i , which is an independent, identically distributed realization from the joint distribution $f(X_S)$ with mean vector $(\bar{X}_{S1}, \bar{X}_{S2}, \dots)$ and covariance matrix Σ . Assume the studies and the target population sample may have different mean vectors, but their covariance matrices are the same. Next suppose the outcome is a function of the observed covariates and treatment, $h(A, X) + \epsilon$ (a zero-mean random error), which is the same in each study and in the target population sample. This function is assumed to be linear below, but in general can take any form. These assumptions are restrictive and arguably untenable in some situations. It is necessary, however, to make rather strong assumptions when using limited aggregate information to inform an IPD analysis. These assumptions may be reasonable for many applications; next, we discuss ways to relax them.

Aggregate-matched synthetic IPD

We generate AMSIPD in 2 steps: covariates, then outcomes. We propose 2 approaches for generating synthetic values: a simpler version that assumes homogeneity among the meta-analysis

studies and a version that allows heterogeneity. [Appendix S1](#) describes the process algorithmically.

Step 1: Generate aggregate-matched covariates

Assuming homogeneity, we begin by estimating the covariate distribution using all available IPD—meta-analysis studies with IPD and the target population sample—using nonparametric density estimation, specifically the R package *np*.¹³ For low-dimensional problems, this flexibly estimates the joint distribution of both categorical and continuous covariates using a product kernel: for continuous variables, a Gaussian kernel, and for categorical variables, a kernel described by Shalizi.¹⁴ We then sample “new” units from the estimated joint distribution as follows.¹⁴

First, we sample (uniformly, with replacement) a unit (with covariates) from the combined data set of studies with IPD and the target population sample, then we sample a synthetic unit from the estimated density using the appropriate kernel. For continuous variables, we sample the covariate for the synthetic unit by randomly drawing from a Gaussian distribution with the mean equal to the observed covariate value for the sampled unit and SD equal to the bandwidth estimated in the nonparametric density procedure. For binary variables, the estimated bandwidth is the probability of keeping the observed covariate value of the sampled unit rather than switching to the other value. Because the density estimate uses a product kernel, this sampling can be done marginally for each variable after sampling a unit from the combined data set of studies with IPD and the target population sample. For each unit randomly sampled from the combined data set of studies and IPD, this procedure generates a “synthetic” unit sampled from the joint distribution centered at the “true” sampled unit’s covariate values.

Assuming heterogeneity, the process is as just described, but instead of estimating the distribution of covariates using all available IPD, we first estimate the joint distribution of covariates separately for each study, using nonparametric density estimation; then, when generating synthetic data, for each unit we first sample randomly a study from the studies with IPD (eg, study 2) and then use the above process with only that study. Thus, for each draw in a sample of size n , we sample a study (with replacement), then sample a new unit from that study’s estimated density. This approach captures observed heterogeneity between studies and weakens the assumption that all studies have the same covariance matrix. However, the smaller sample sizes could lead to less-stable estimates of the covariate distribution. In the data analysis example, we assume homogeneity for simplicity, akin to the weaker conditional homogeneity of Dahabreh et al.^{4,5}; in the simulation study, we assume heterogeneity.

Although aggregate-only studies do not have individual-level information, our synthetic data must be consistent with the data we do have. At the study and arm levels, we have counts for categorical covariates and means and SDs for continuous covariates. Counts can be matched directly using rejection sampling: keep only individuals with values of synthetic categorical covariates that are needed to match observed counts, rejecting those with values that are already fully matched. One, instead, could manually change the bandwidth in the density estimation so the expectation of the synthetic counts equals the observed counts. Rejection sampling ensures that the synthetic counts match the observed counts exactly and is easy to understand, but it becomes increasingly inefficient with more categorical covariates. Changing the bandwidth implicitly accounts for sampling variability in the study samples and is more flexible but can be difficult to understand. Continuous covariates are recentered and rescaled

so the synthetic means and SDs match the observed means and SDs (although changing the bandwidth could also be used here). Treatment is then assigned randomly to the synthetic participants as in the original trial. Because we assume treatment assignment is random and independent of covariates, we match covariates to aggregate information at the study level rather than the arm level.

Step 2: Generate aggregate-matched outcomes

Once covariates have been generated, outcomes are generated using available data about the relationship among covariates, treatment, and outcome. This step can use only studies with IPD (covariates and outcome); the target population sample generally cannot be used. The simplest approach is to fit a linear model regressing the outcome on the treatment, covariates, and treatment-covariate interactions, as follows:

$$Y = \beta_0 + \alpha A + \beta X + \gamma AX + \epsilon,$$

estimating β_0 , α , β , and γ using least squares. Then we predict the outcome for each synthetic individual using this fitted model and the synthetic covariates/treatment from step 1, where variables with “hat” accents indicate estimates:

$$\hat{Y}_{\text{AMSIPD}} = \hat{\beta}_0 + \hat{\alpha}\hat{A}_{\text{AMSIPD}} + \hat{\beta}\hat{X}_{\text{AMSIPD}} + \hat{\gamma}\hat{A}\hat{X}_{\text{AMSIPD}}.$$

Nonparametric regression or random forests can be used to estimate the relationship between the outcome and the covariates/treatment; outcomes based on the estimated relationship are predicted the same way.

Recalling that we want the synthetic data to match actual aggregate data, a practical concern is that the synthetic outcomes can be more variable than the observed aggregate outcomes (here, we focus on the arm level within studies, denoted by a and s):

$$\text{Var}(\hat{Y}_{\text{AMSIPD},s,a}) > \text{Var}(Y_{\text{obs},s,a}).$$

When this happens, one cannot simply rescale as in step 1, because such a change could break the estimated relationship between outcomes and covariates/treatment, and one cannot add random variation to the synthetic outcomes, because the latter are already too variable. A naïve solution to this problem is to repeat the entire process, from step 1, until the synthetic outcomes are not too variable, then add noise to match the variability of the synthetic outcomes to the observed outcomes. This solution is computationally inefficient and may never produce suitable synthetic outcomes. A more disciplined approach fits the model as a quadratic programming problem, minimizing the sum of squared errors for the outcome subject to a constraint that the synthetic outcome variance is no greater than the observed outcome variance for each study-arm combination:

$$\text{minimize}_{\beta_0, \alpha, \beta, \gamma} [Y - (\beta_0 + \alpha A + \beta X + \gamma AX)]$$

$$\text{s.t.} \quad \text{Var}(\hat{Y}_{\text{AMSIPD},s,a}) \leq \text{Var}(Y_{\text{obs},s,a}).$$

When the simple linear model approach predicts outcomes that are not too variable, it is equivalent to the quadratic programming approach. When the linear model predicts outcomes that are too variable, quadratic programming fixes the problem. This does not fix excessive variability when using other model-fitting methods, like random forests.

Table 1. Simulation study: true covariate mean vectors.^a

Scenario	S = 0	S = 1	S = 2	S = 3	S = 4	S = 5
1	(0, 0)	(-1.5, 1.5)	(-2, -2)	(2, -1)	(0, -1.5)	(1, 1.5)
2	(0, 0)	(-3, 3)	(-3.5, -2)	(3.5, 0)	(0, -3.5)	(1, 3.5)
3	(3, 0)	(0, 4)	(0, -4)	(0, 0)	(0, -2)	(0, 2)
4	(0, 0)	(2, -2)	(-2, 2)	(2, 2)	(0, 1)	(0, -1)

^aRows in this table correspond to scenarios, columns to the target population ($S = 0$) and the meta-analysis studies ($S > 0$). These 4 scenarios are depicted in the 4 panels of [Figure 1](#).

Step 3: Repeat steps 1 and 2 to account for variability in the process

The AMSIPD process has inherent randomness, so it should be run repeatedly, like multiple imputation. The analyses we report here use 5 runs, but other numbers of runs can be used. We estimate overall ATEs as the mean across runs of the estimated ATE, and we estimate the variance of estimated treatment effects using Rubin's rules.¹⁵

Having generated synthetic covariates and outcomes for each participant in aggregate-data-only studies, we have an IPD data set for each meta-analysis study and can estimate ATEs using any CIMA method that requires IPD (eg, the outcome model or inverse probability weighting approaches of Dahabreh et al.⁴⁻⁶).

Evaluation of the AMSIPD methods: case study

We evaluate the proposed methods using a case study of real-time continuous glucose monitoring,¹⁶ a leave-1-out evaluation in which each study is treated in turn as the target population sample, and the other studies are meta-analyzed to estimate the ATE in that target population. The continuous outcome is glycated hemoglobin at follow-up, with covariates baseline glycated hemoglobin, age, and binarized sex.⁶ Performance is evaluated using the average absolute difference metric described by Rott et al.,⁶ the absolute difference between the transported estimate and the observed ATE for each target, averaged across the targets. The standardized absolute difference, which adjusts the absolute difference for sample size, ranked methods similarly and by similar margins as average absolute difference in the Rott et al. article.⁶

We treat half of the 8 studies (labeled 2, 3, 4, and 5) as if they were aggregate only instead of IPD when they are used as meta-analysis studies. That is, we use only counts or means and SDs for covariates for each treatment group within the study. If a study represents the target population sample in the leave-1-out exercise, it is treated as having IPD and is excluded from the aggregated studies if necessary.

We compare the new methods to “full IPD” and “IPD only” methods, in which estimation methods are applied (respectively) to IPD from all studies or excluding the now-aggregated studies. The full IPD comparator should be an upper bound on AMSIPD's performance, whereas IPD only reflects a choice between excluding aggregate-only studies or using our approach. We considered these estimation methods, which are among the most commonly used estimators in causal inference and thus interesting (our earlier article⁶ gives detailed descriptions): outcome modeling (OM), inverse probability weighting (IPW) using logistic regression for the propensity score, IPW using nonparametric density estimation for the propensity score (np IPW), and Hajek-normalized versions of IPW and np IPW.⁴⁻⁶

Evaluation of AMSIPD methods: simulation study

In addition to the case study, we also present simulation results. For all simulation scenarios, we have 5 meta-analysis studies and an IPD target population sample. As described earlier in this article, we label these data sources $S = 0$ for the target population sample and $S \in \{1, 2, 3, 4, 5\}$ for the meta-analysis studies. In all scenarios, studies 4 and 5 are treated as if they are aggregate only (except for full IPD). We simulate outcomes under 2 settings: first, a linear relationship with treatment and covariates (and their interactions), which should favor the 2 IPD methods (full IPD and IPD only), especially when using the OM method, because the assumed model matches the data generating process; and second, a specification that adds quadratic effect modification. [Appendix S2](#) shows the specific relationships. We consider 4 sample size scenarios, in all of which the target population samples and studies 1 and 2 have a sample size of 100. The scenarios differ in the number of studies with a sample size of 400: in the first, studies 3, 4, and 5, $n = 400$; in the second, studies 3 and 4 do; in the third scenario, only study 3 does; and in the fourth all studies have a sample size of 100.

The overlap between the meta-analysis studies and the target population sample should affect performance of the AMSIPD methods relative to traditional meta-analysis methods and the IPD CIMA methods. Thus, we also consider 4 overlap scenarios, shown in [Table 1](#) and [Figure 1](#). Each data source has 2 covariates (X_1, X_2) sampled from a multivariate normal distribution; the mean vector is reported in [Table 1](#). The target population sample is generated with the covariance matrix $\begin{bmatrix} 2 & 0.25 \\ 0.25 & 2 \end{bmatrix}$, and the meta-analysis studies are generated with the covariance matrix $\begin{bmatrix} 1 & 0.25 \\ 0.25 & 1 \end{bmatrix}$. [Figure 1](#) shows simulated data from each scenario, which have rationales as follows.

Scenario 1: Good overlap with union of studies. The meta-analysis studies have minor overlap with each other, but their union covers the target population. In this base case, most methods will likely do well when all studies are included, but losing some studies could be problematic ([Figure 1A](#)).

Scenario 2: Minimal overlap with target population, requiring interpolation of studies. This resembles scenario 1 in that the target population falls in the center of the union of meta-analysis studies, but now most of the target population does not actually overlap with those studies. This violates the positivity assumption and shows how this violation affects estimates for the competing methods ([Figure 1B](#)).

Scenario 3: Target population has minimal overlap. The meta-analysis studies have significant overlap with one another but little with the target population. All methods are extrapolating and expected to perform poorly: the positivity assumption is severely violated ([Figure 1C](#)).

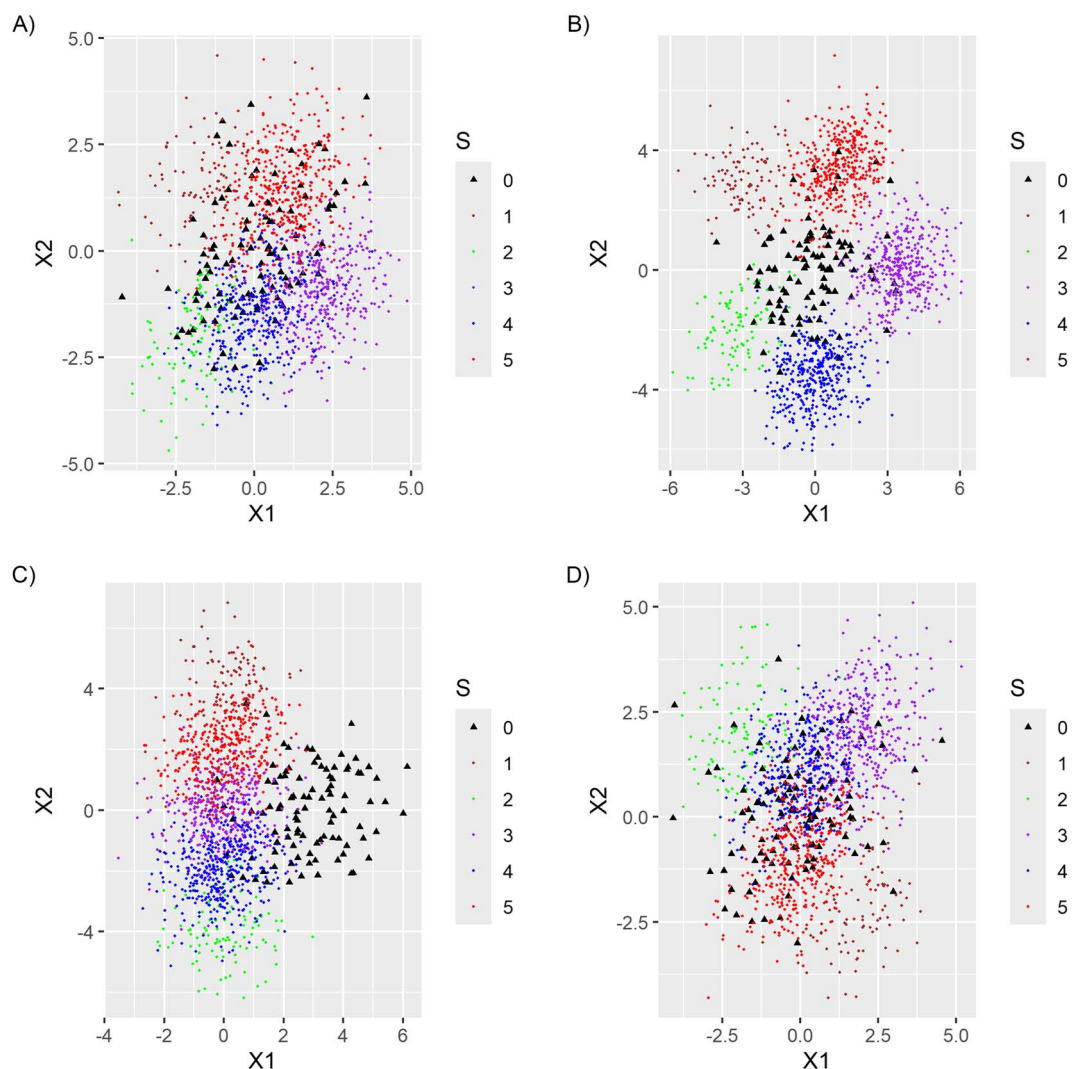


Figure 1. Example data from the 4 simulation scenarios, as in the panel captions. A) Scenario 1: good overlap with union of studies. B) Scenario 2: minimal overlap with target population, requiring interpolation of studies. C) Scenario 3: target population has minimal overlap. D) Scenario 4: strongest overlap with aggregate-only studies.

Scenario 4: Strongest overlap with aggregate-only studies. The target population has minor overlap with the 3 IPD studies and has strong overlap with the 2 aggregate-only studies. We expect that losing the 2 studies with strong overlap will hamper the usual CIMA methods, whereas AMSIPD will restore them (Figure 1D).

These scenarios are useful as either an idealized but reasonable depiction of potential data sets or for pushing some methods toward their breaking points, to see when 1 method is more likely to outperform others.

Results

Case study

For the case study, Table 2 compares the AMSIPD methods with methods using IPD. For OM estimators, AMSIPD performs almost as well as using full IPD from all studies, and outperforms IPD-only method by about 25%. For weighting estimators, AMSIPD outperforms full IPD and IPD only across the board, sometimes by nearly an order of magnitude. Even for the weighting estimator that performs best on full IPD (ie, Hajek-normalized np IPW), AMSIPD performs 20% to 30% better. The Hajek-normalized IPW estimates using AMSIPD outperform even the OM estimator. The performance improvement for basic IPW, which estimates the

propensity score using logistic regression, is astronomical, about 80%. Although striking, perhaps this is not surprising: weighting methods have well-known instability due to large variability in the weights,¹⁷ especially when the positivity assumption is violated. This motivates weighting “fixes” like trimming and truncation, which improve stability but change the target population.^{18,19} As discussed by Rott et al.,⁶ this case study has a clear positivity problem. Because the AMSIPD process effectively shrinks synthetic individuals toward the observed aggregates, it should reduce the chance that a few individuals receive exaggerated weights and thereby improve performance. The OM estimate is not as sensitive to a few individuals and, as discussed above, AMSIPD OM estimates perform like full IPD. Thus, although the stark improvement from using AMSIPD with weighting methods may be surprising and is likely more extreme than it would be in many applied cases, it is real and explainable. Because the quadratic programming approach performed well in this case study and has desirable properties, we used it in the simulation study.

Simulation study

Table 3 summarizes results for simulated data with linear effect modification. The OM estimates are nearly unbiased in all

Table 2. Case study: comparison of aggregate-matched synthetic individual participant data and observed individual participant data for the real-time continuous glucose monitoring data.^a

	OM	IPW	IPW-h ^b	np IPW	np IPW-h ^b
Full IPD ^c	0.236	2.281	0.317	0.593	0.264
IPD only ^d	0.346	1.085	0.390	0.590	0.334
AMSIPD LM ^e	0.245	0.495	0.190	0.305	0.202
AMSIPD npreg	0.238	0.434	0.178	0.315	0.196
AMSIPD QP ^f	0.245	0.448	0.179	0.291	0.189

Abbreviations: AMSIPD, aggregate-matched synthetic individual participant data; IPW, inverse probability weighting; LM, linear model; np IPW, inverse probability weighting using nonparametric conditional density estimation; npreg, nonparametric regression; OM, outcome model; QP, quadratic programming. ^aThis table describes a leave-1-out evaluation in which each study in the continuous glucose monitoring data set is used, in turn, as the sample from the target population, and its actual (estimated) treatment effect is compared with a treatment effect estimate obtained by meta-analyzing the remaining studies. Rows in the table correspond to AMSIPD and its comparators, with full IPD being the idealized case in which all studies have IPD. Columns correspond to causally interpretable meta-analysis estimation methods, any of which can be used for any row. Each table entry is the average absolute difference for the leave-1-out evaluation when the column method is applied to the row.

^bThe -h suffix indicates Hajek normalization.

^cUsing the IPD from all studies.

^dUsing IPD only from those studies assumed to have IPD in the exercise.

^eA basic linear model was used to predict outcomes.

^fA linear model with the quadratic programming approach.

scenarios for full IPD and IPD only. The OM estimators are expected to perform better than weighting estimators because the simulated outcomes are linearly related to the predictors, like the fitted outcome model. We report in subsequent paragraphs results from simulations with nonlinear effect modification. The AMSIPD had both encouraging and discouraging results. Their OM estimates are substantially worse than other methods', with large biases in several scenarios. It is not obvious this pattern will remain when outcomes are generated using other models. For weighting methods, however, AMSIPD performs at least as well as the IPD-only method in most settings. In scenarios 2 and 3, AMSIPD outperforms IPD only for any weighting method and sample size. In scenario 1, AMSIPD using nonparametric density estimation outperforms IPD only, but IPD only using logistic regression outperforms AMSIPD. In scenario 4, that trend is largely flipped, although for some sample sizes either AMSIPD or IPD only is a clear winner.

Table 4 summarizes results under quadratic effect modification. Because all estimation methods and synthetic data generation use misspecified models, we expect all methods to perform poorly, and they generally do. Aggregate-matched synthetic IPD are competitive with IPD only and full IPD, and perform favorably in scenario 3's extreme case.

Figures S1-S8 show graphical results for both types of effect modification. In summary, in a situation favorable to IPD only, when the true data-generating model is linear in the effect modifiers, the AMSIPD was somewhat worse than IPD only for OM estimators, but for weighting methods, it was, at worst, competitive and often was superior. Combined with the results from the case study, which was not designed to be advantageous to IPD only, these results suggest that AMSIPD is a reasonable alternative to IPD only and is worth further exploration.

Discussion

The case study and simulation study suggest that AMSIPD warrants further study and could address a key problem facing CIMA. Currently, investigators have 2 choices: traditional meta-analysis or CIMA. Traditional methods lack CIMA's causal foundation, and many use only aggregate data, losing the advantages of analyzing IPD. Although CIMA solves those problems, its need for IPD can produce bias when studies with and without IPD differ systematically. Aggregate-matched synthetic IPD allows all studies to be included, enhancing CIMA's appeal. The former still has clear lim-

itations (eg, assuming all studies have the same covariates). More broadly, the assumptions made here are quite strong and may not be satisfied in many applications. Investigating the method's robustness to violations of the stronger assumptions is important future work. Furthermore, sensitivity analysis methods to assess robustness to violations of this assumption are possible, but they are outside this article's scope. A recent approach is amenable to our proposed methods.¹² Beyond the causal assumptions, AMSIPD requires 2 key assumptions: 1 about the covariance of covariates and another about the outcome mean function. Our simulation study provides some evidence about the effects of violating them; here, we offer some speculation to supplement the simulation results. We require that the covariance of the covariates is either the same in all studies (homogeneous assumption) and in the target population, or the target population's covariance is a mixture of the covariances of the meta-analysis studies with IPD (heterogeneity assumption). We speculate that failure of this assumption has only modest effects unless the failure is gross, because this is used only to simulate covariates for individuals for aggregate-only studies. The second assumption requires that the true conditional mean outcome as a function of covariates and treatments is the same for all studies and is specified correctly. This assumption is likely more critical than the assumption about covariate covariance. However, again, we speculate that the effect of violating it is likely to be modest as long as the misspecification is not severe, as has been noted for similar assumption violations in the causal literature.²⁰ Although this method can be used directly (eg, for binary and count outcomes), further work is needed for more complex settings like time-to-event outcomes. Despite these shortcomings, however, AMSIPD is a good starting point for combining aggregate and IPD for CIMA.

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Conflict of interest

The authors declare no conflicts of interest.

Data availability

The data that support the findings of this study are potentially available on request. The data are not publicly available, due to privacy or ethical restrictions.

Table 3. Simulation study: data with linear effect modification.^a

Sample size, no.	Scenario no.	Full IPD ^b				IPD only ^c				AMSIPD QP ^d						
		OM	IPW	IPW-h ^e	np IPW	np IPW-h ^e	OM	IPW	IPW-h ^e	np IPW	np IPW-h ^e	OM	IPW	IPW-h ^e	np IPW	np IPW-h ^e
1	1	-0.014	-0.017	-0.017	0.202	0.199	-0.017	-0.082	-0.081	-0.423	-0.483	-0.265	-0.254	-0.254	-0.113	-0.112
1	2	-0.018	-0.125	-0.124	0.146	0.225	-0.017	-0.506	-0.497	-0.119	-0.211	-0.130	-0.227	-0.225	-0.002	0.001
1	3	0.014	-3.230	-1.981	-4.147	-2.300	0.012	-3.448	-1.850	-4.565	-2.155	2.577	-2.019	-0.080	-3.705	-1.161
1	4	0.005	-0.050	-0.049	0.207	0.204	0.002	-0.994	-0.951	0.183	0.237	-0.094	-0.169	-0.168	0.264	0.262
2	1	-0.022	0.006	0.005	0.004	0.004	-0.018	-0.089	-0.089	-0.435	-0.499	-0.283	-0.225	-0.224	-0.419	-0.424
2	2	-0.014	0.012	0.012	0.031	0.049	-0.012	-0.481	-0.472	-0.129	-0.248	-0.155	-0.098	-0.097	-0.091	-0.178
2	3	-0.010	-3.358	-2.012	-4.224	-2.472	-0.009	-3.533	-1.797	-4.537	-2.139	1.992	-2.398	-0.578	-3.919	-1.695
2	4	0.005	-0.186	-0.183	0.242	0.242	0.005	-1.000	-0.955	0.189	0.240	-0.011	-0.212	-0.209	0.451	0.462
3	1	0.015	-0.009	-0.009	0.019	0.020	0.012	-0.070	-0.069	-0.427	-0.493	-0.114	-0.132	-0.131	-0.243	-0.245
3	2	0.020	-0.254	-0.251	0.154	0.259	0.019	-0.473	-0.464	-0.066	-0.121	-0.014	-0.287	-0.283	0.005	0.017
3	3	0.001	-3.306	-1.993	-4.223	-2.147	0.000	-3.288	-1.761	-4.508	-2.115	1.109	-2.709	-1.115	-4.027	-1.667
3	4	0.003	-0.348	-0.341	0.151	0.156	0.003	-0.954	-0.915	0.194	0.246	-0.054	-0.426	-0.419	0.183	0.193
4	1	-0.008	0.045	0.044	-0.043	-0.040	-0.005	0.075	0.074	-0.565	-0.661	-0.042	0.003	0.002	-0.213	-0.207
4	2	-0.022	0.028	0.028	-0.002	-0.004	-0.022	0.551	0.539	-0.245	-0.624	0.021	0.079	0.078	-0.239	-0.497
4	3	0.015	-3.263	-2.409	-4.544	-2.596	0.014	-3.441	-2.058	-4.809	-2.392	1.829	-2.440	-0.883	-4.246	-1.751
4	4	0.014	0.031	0.031	0.074	0.076	0.009	0.006	0.006	0.003	0.006	-0.112	-0.123	-0.124	-0.012	-0.014

Abbreviations: AMSIPD, aggregate-matched synthetic individual participant data; IPW, inverse probability weighting using nonparametric conditional density estimation; npreg, nonparametric regression; OM, outcome model; QP, quadratic programming.

^aThis table shows the bias for idealized full IPD, and 2 realistic competitors: using studies with IPD only and AMSIPD. For each competitor, we show the bias for 5 estimation methods.

^bFull IPD indicates using the IPD from all studies.

^cIPD only indicates using IPD only from those studies assumed to have IPD in the exercise.

^dA linear model with the quadratic programming approach.

^eThe -h suffix indicates Hajek normalization.

Table 4. Simulation study: data with quadratic effect modification.^a

Sample size, no	Scenario no.	Full IPD ^b				IPD only ^c				AMSIPD QP ^d						
		OM	IPW	IPW-h ^e	Np IPW	Np IPW-h ^e	OM	IPW	IPW-h ^e	Np IPW	Np IPW-h ^e	OM	IPW	IPW-h ^e	Np IPW	Np IPW-h ^e
1	1	1.987	1.912	1.923	2.164	2.182	1.988	1.800	1.845	1.306	0.922	1.701	1.688	1.693	1.820	1.856
1	2	1.982	1.845	1.853	1.060	-0.203	1.987	1.443	1.480	1.443	-1.351	1.842	1.728	1.737	2.027	0.425
1	3	2.017	-3.923	-4.445	-2.891	-3.725	2.009	-3.208	-3.453	-2.408	-2.992	5.171	-0.160	0.706	-1.222	-0.330
1	4	2.009	1.885	1.887	2.213	2.199	2.011	0.782	0.901	1.524	0.914	1.857	1.747	1.749	2.154	2.117
2	1	1.983	1.892	1.912	1.961	1.941	1.991	1.796	1.845	1.303	0.920	1.724	1.710	1.735	1.551	1.528
2	2	1.995	1.979	2.001	1.235	-0.168	1.998	1.474	1.510	1.403	-1.384	1.844	1.871	1.894	1.873	0.142
2	3	2.002	-3.743	-4.320	-2.976	-3.864	2.011	-3.140	-3.367	-2.376	-2.940	4.447	-1.013	-0.443	-1.660	-1.286
2	4	2.006	1.707	1.725	2.197	2.155	2.005	0.764	0.887	1.489	0.872	1.934	1.668	1.689	2.320	2.244
3	1	2.010	1.911	1.934	1.971	1.944	2.013	1.809	1.860	1.314	0.921	1.881	1.804	1.827	1.634	1.605
3	2	2.024	1.717	1.736	1.464	0.097	2.024	1.481	1.516	1.492	-1.271	1.978	1.678	1.697	1.812	0.155
3	3	1.998	-3.515	-3.927	-2.570	-3.201	2.003	-3.058	-3.406	-2.387	-2.981	3.326	-1.772	-1.640	-1.912	-1.796
3	4	2.002	1.515	1.543	2.101	2.002	2.002	0.816	0.928	1.506	0.907	1.916	1.435	1.464	1.964	1.806
4	1	1.992	2.023	2.026	1.941	1.990	1.995	1.980	1.987	1.278	0.836	1.964	1.997	1.999	1.704	1.750
4	2	1.976	2.015	2.020	1.454	-0.028	1.974	2.495	2.521	1.777	-1.238	2.024	2.075	2.080	2.014	0.079
4	3	2.008	-3.715	-4.051	-2.524	-3.191	1.997	-3.035	-3.128	-2.229	-2.650	4.170	-1.027	-0.486	-1.615	-1.047
4	4	2.028	1.991	1.989	2.142	2.100	2.031	1.921	1.913	1.635	0.926	1.889	1.847	1.842	1.942	1.821

Abbreviations: aggregate-matched synthetic individual participant data; IPW, inverse probability weighting; np IPW, inverse probability weighting using nonparametric conditional density estimation.

^aThis table shows the bias for idealized full IPD and 2 realistic competitors: using studies with IPD only and AMSIPD. For each competitor, we show the bias for 5 estimation methods.
^bUsing the IPD from all studies.
^cUsing IPD only from those studies assumed to have IPD in the exercise.
^dThe synthetic IPD process with the quadratic programming approach.
^eThe -h suffix indicates Hajek normalization.

Supplementary material

Supplementary material is available at the American Journal of Epidemiology online.

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